# Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Distance and midpoint formulas

Find the distance between $(2, -1)$ and $(10, 5)$.

**2** [Foundational] • Target: Distance and midpoint formulas

Find the midpoint of the segment connecting $(-4, 7)$ and $(6, -3)$.

**3** [Foundational] • Target: Slope and equations of lines

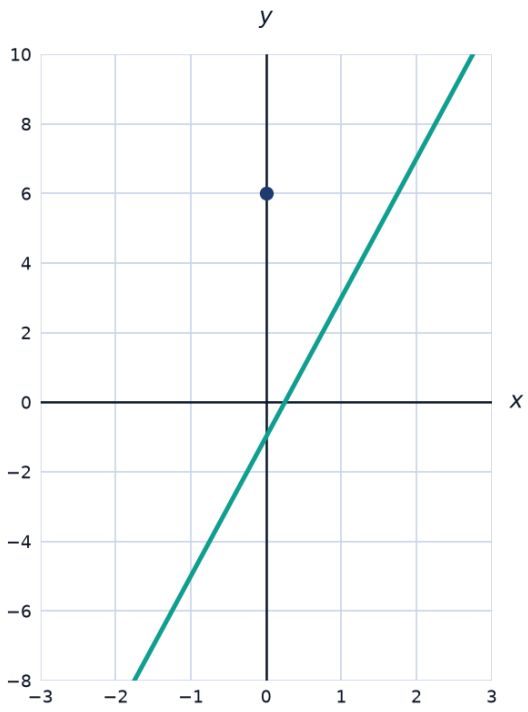
Find the slope of the line through $(1, 4)$ and $(5, -8)$.

**4** [Foundational] • Target: Slope and equations of lines

Write the equation of the line through $(2, -3)$ with slope $\frac{1}{2}$, in slope-intercept form.

**5** [Foundational] • Target: Parallel and perpendicular lines

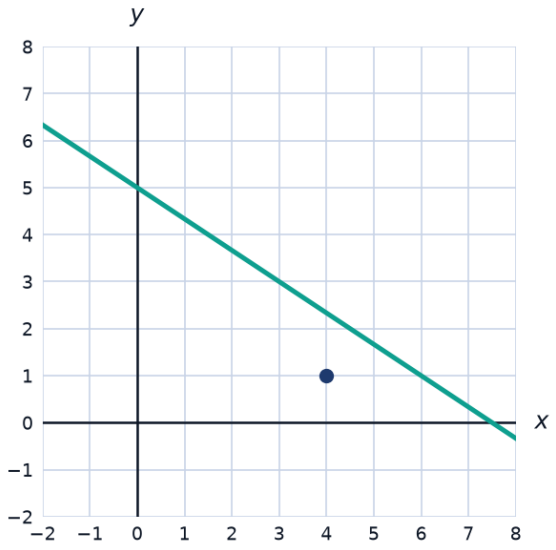
Find the equation of the line parallel to $y = 4x - 1$ passing through $(0, 6)$.




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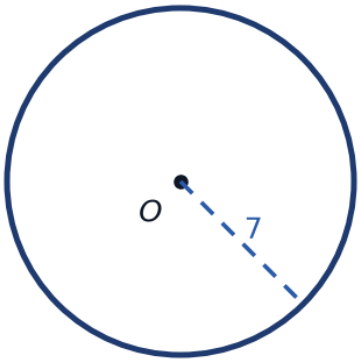
**6** [Foundational] • Target: Parallel and perpendicular lines

Find the equation of the line perpendicular to $y = -\frac{2}{3}x + 5$ passing through $(4, 1)$.




**7** [Foundational] • Target: The equation of a circle

Write the equation of a circle with center $(3, -2)$ and radius $7$.




## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

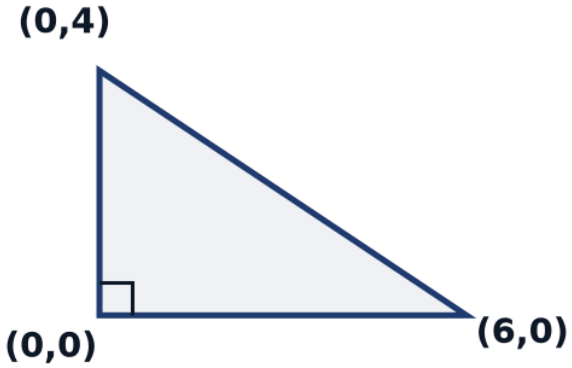
**8** [Intermediate] • Target: Reflections, translations, and symmetry

Find the coordinates of the point $(5, -3)$ after a reflection over the $x$-axis, then after a reflection over the $y$-axis.

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**9** [Intermediate] • Target: Finding area using the coordinate plane

Find the area of the triangle with vertices $(0, 0)$, $(6, 0)$, and $(0, 4)$.




**10** [Intermediate] • Target: Finding area using the coordinate plane

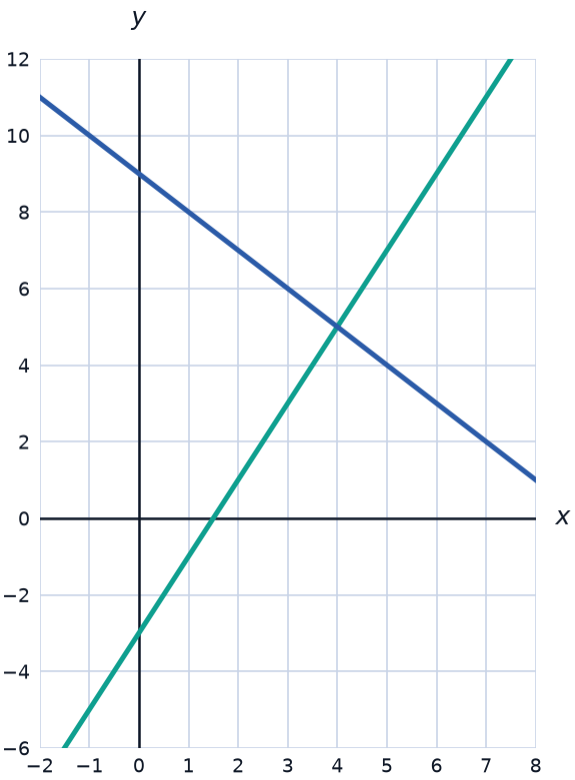
A rectangle has vertices $(1, 1)$, $(1, 5)$, $(7, 5)$, and $(7, 1)$. Find its area and perimeter.

**11** [Intermediate] • Target: Determining if points are collinear

Determine whether the points $(1, 2)$, $(3, 6)$, and $(5, 10)$ are collinear, by comparing slopes.

**12** [Intermediate] • Target: Finding the intersection of two lines

Find the point of intersection of $y = 2x - 3$ and $y = -x + 9$.




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**13** **[Intermediate]** • **Target: Combining coordinate geometry concepts**

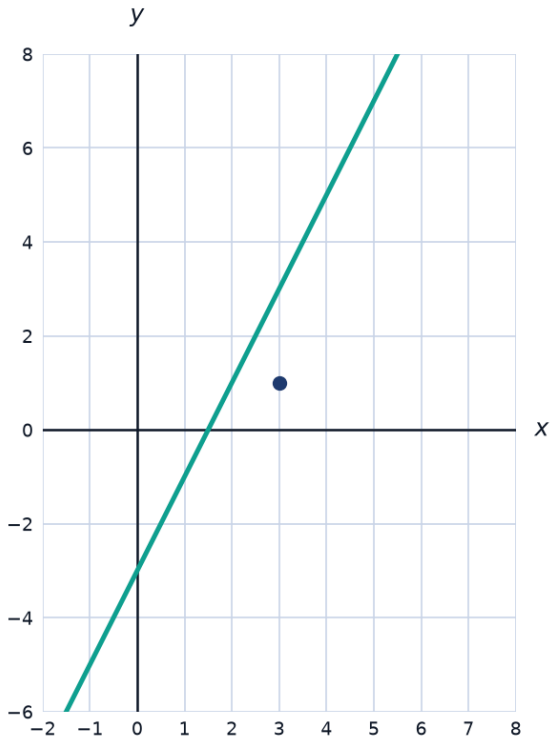
This question has two parts. Triangle $ABC$ has vertices $A(0, 0)$, $B(4, 0)$, and $C(2, 6)$. (a) Find the length of each side of the triangle. (b) Determine whether the triangle is isosceles, and justify your answer.




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**14** [Intermediate] • Target: Graphing inequalities in the coordinate plane

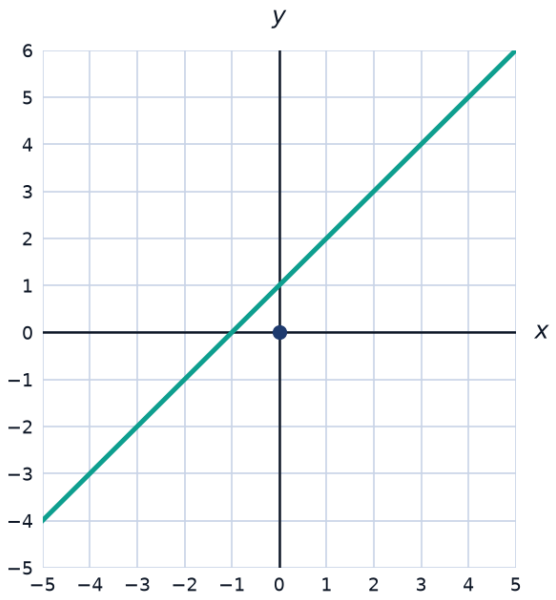
Determine whether the point $(3, 1)$ satisfies the inequality $y < 2x - 3$.




## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Graphing inequalities in the coordinate plane • Cross-Lesson: 5.01 Plane Geometry

A region is defined by $y \geq x + 1$. Determine whether $(0, 0)$ lies in this region.




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**16** [Challenging] • Target: The equation of a parabola in the coordinate plane • Cross-Lesson: 6.01 Trigonometry

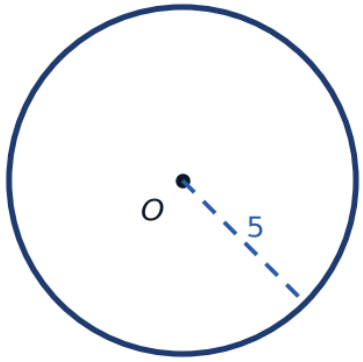
Write the equation of a parabola with vertex $(2, -3)$ that passes through $(4, 5)$.

**17** [Challenging] • Target: Finding points that divide a segment in a given ratio • Cross-Lesson: 7.01 Statistics & Probability

Find the point that divides the segment from $(1, 2)$ to $(9, 10)$ in a ratio of $1:3$ (closer to the first point).

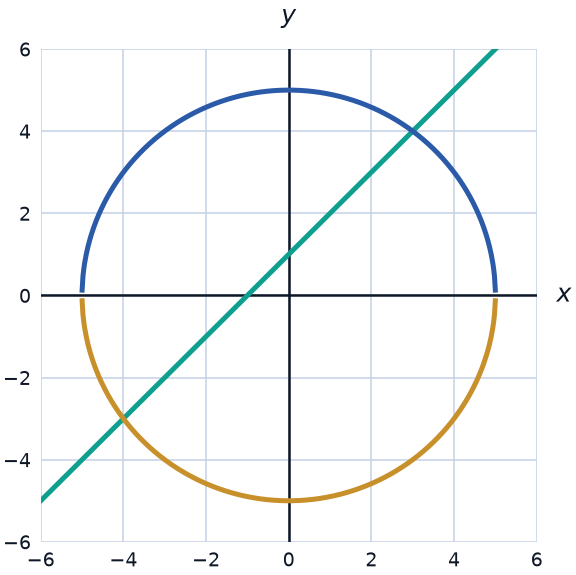
**18** [Challenging] • Target: A multi-part problem combining lines and circles • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

This question has two parts. A circle has equation $(x-2)^2 + (y+1)^2 = 25$. (a) State the center and radius. (b) Determine whether the point $(5, 3)$ lies inside, on, or outside the circle, by computing its distance from the center.




**19** [Challenging] • Target: Systems of a line and a circle • Cross-Lesson: 1.01 Numbers, Percents & Ratios

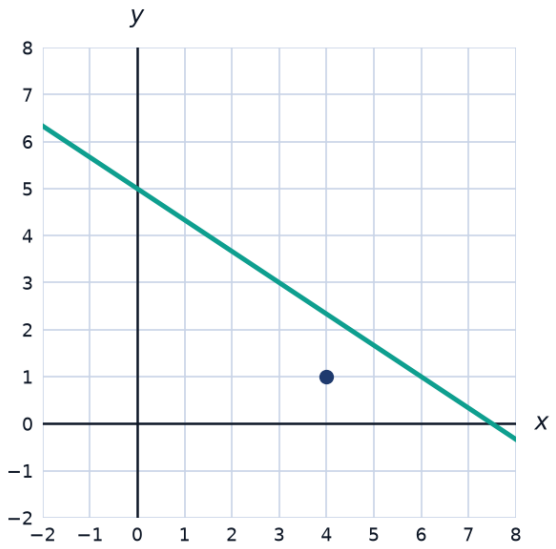
Find the points of intersection of the line $y = x + 1$ and the circle $x^2 + y^2 = 25$.




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**20** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the equation of the line perpendicular to $y = -\frac{2}{3}x + 5$ passing through $(4, 1)$.




## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Distance and midpoint formulas

Find the distance between $(8, 8)$ and $(18, 32)$.

**a.** 27

**b.** 34

**c.** 25

**d.** 26

**22** [Foundational] • Target: Slope and equations of lines

Find the distance between $(-2, 7)$ and $(-29, -29)$.

**a.** 63

**b.** 46

**c.** 45

**d.** 44

**23** [Foundational] • Target: Parallel and perpendicular lines

Find the distance between $(0, 2)$ and $(8, 17)$.

**a.** 16

**b.** 18

**c.** 17

**d.** 23

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**24** [Foundational] • Target: The equation of a circle

Find the distance between $(-2, 4)$ and $(5, -20)$.

**a.** 26

**b.** 24

**c.** 31

**d.** 25

**25** [Foundational] • Target: Reflections, translations, and symmetry

Find the distance between $(6, -3)$ and $(-12, -27)$.

**a.** 29

**b.** 42

**c.** 30

**d.** 31

**26** [Foundational] • Target: Finding area using the coordinate plane

Find the distance between $(-7, -6)$ and $(17, -51)$.

**a.** 52

**b.** 50

**c.** 51

**d.** 69

**27** [Foundational] • Target: Determining if points are collinear

Find the distance between $(3, -5)$ and $(-17, 43)$.

**a.** 52

**b.** 53

**c.** 68

**d.** 51

**28** [Foundational] • Target: Finding the intersection of two lines

Find the distance between $(8, -5)$ and $(-19, -125)$.

**a.** 122

**b.** 123

**c.** 124

**d.** 147

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: Combining coordinate geometry concepts

Find the midpoint of the segment connecting $(9, -5)$ and $(1, 11)$.

**a.** $(5, 4)$

**b.** $(6, 3)$

**c.** $(5, 3)$

**d.** $(-5, 3)$

**30** [Intermediate] • Target: Graphing inequalities in the coordinate plane

Find the midpoint of the segment connecting $(-6, 0)$ and $(4, 12)$.

**a.** $(1, 6)$

**b.** $(0, 6)$

**c.** $(-1, 6)$

**d.** $(-1, 7)$

**31** [Intermediate] • Target: The equation of a parabola in the coordinate plane

Find the midpoint of the segment connecting $(2, 8)$ and $(10, -4)$.

**a.** $(6, 3)$

**b.** $(6, 2)$

**c.** $(7, 2)$

**d.** $(-6, 2)$

**32** [Intermediate] • Target: Finding points that divide a segment in a given ratio

Find the midpoint of the segment connecting $(7, 3)$ and $(-11, -11)$.

**a.** $(-2, -3)$

**b.** $(-1, -4)$

**c.** $(2, -4)$

**d.** $(-2, -4)$

**33** [Intermediate] • Target: A multi-part problem combining lines and circles

Find the midpoint of the segment connecting $(8, 1)$ and $(0, -7)$.

**a.** $(5, -3)$

**b.** $(4, -3)$

**c.** $(4, -2)$

**d.** $(-4, -3)$

**34** [Intermediate] • Target: Systems of a line and a circle

Find the midpoint of the segment connecting $(-7, -3)$ and $(3, 1)$.

**a.** $(-2, 0)$

**b.** $(-2, -1)$

**c.** $(2, -1)$

**d.** $(-1, -1)$

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**35** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the midpoint of the segment connecting $(12, 9)$ and $(-4, 11)$.

**a.** $(-4, 10)$

**b.** $(4, 11)$

**c.** $(5, 10)$

**d.** $(4, 10)$

**36** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the radius of the circle $x^2 + y^2 + 16x - 18y + 81 = 0$, by completing the square.

**a.** 8

**b.** 7

**c.** 64

**d.** 9

**37** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the radius of the circle $x^2 + y^2 - 16y + 15 = 0$, by completing the square.

**a.** 49

**b.** 7

**c.** 8

**d.** 6

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

Find the radius of the circle $x^2 + y^2 - 8x - 16y - 41 = 0$, by completing the square.

**a.** 11

**b.** 121

**c.** 12

**d.** 10

**39** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Find the equation of the line perpendicular to $y = -\frac{1}{4}x$ passing through $(0, 5)$, in slope-intercept form.

**a.** $y = -\frac{1}{4}x + 5$

**b.** $y = 4x + 6$

**c.** $y = -4x + 5$

**d.** $y = 4x + 5$

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**40** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the equation of the line perpendicular to $y = -\frac{2}{3}x$ passing through $(3, -1)$, in slope-intercept form.

**a.** $y = \frac{3}{2}x - \frac{9}{2}$

**b.** $y = -\frac{2}{3}x - \frac{11}{2}$

**c.** $y = -\frac{3}{2}x - \frac{11}{2}$

**d.** $y = \frac{3}{2}x - \frac{11}{2}$

**41** [Challenging] • Target: Distance and midpoint formulas • Cross-Lesson: 3.01 Functions & Graphs

Find the point of intersection of $y = 3x - 17$ and $y = -2x - 2$.

**a.** $(3, -8)$

**b.** $(4, -8)$

**c.** $(-3, -8)$

**d.** $(3, 8)$

**42** [Challenging] • Target: Slope and equations of lines • Cross-Lesson: 5.01 Plane Geometry

Find the point of intersection of $y = x - 12$ and $y = 4x - 39$.

**a.** $(9, 3)$

**b.** $(-9, -3)$

**c.** $(9, -3)$

**d.** $(10, -3)$

**43** [Challenging] • Target: Parallel and perpendicular lines • Cross-Lesson: 6.01 Trigonometry

Find the point of intersection of $y = 5x + 38$ and $y = -3x - 10$.

**a.** $(-5, 8)$

**b.** $(6, 8)$

**c.** $(-6, -8)$

**d.** $(-6, 8)$

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

Which of the following points lies on the line that passes through $(1, 4)$ and has a slope of $-3$?

**a.** $(3, -2)$

**b.** $(2, 2)$

**c.** $(4, -4)$

**d.** $(0, 6)$

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**45** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A circle has the equation $x^2 + y^2 - 6x + 8y = 0$. What is the radius of this circle?

**a.** 4

**b.** 5

**c.** 6

**d.** 10

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